PRAKTIKUM BASIS DATA

MODUL 5

05_Scripting_Agregate_Function

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TP

1. **Jumlah film yang tersedia**
   - **Query**

```
--soal 1
SELECT COUNT(*) AS jumlah_film_tersedia
FROM FILM;
```

   - **Output**

	JUMLAH_FILM_TERSEDIA
1	4

2. **Jumlah film yang sedang diputar (tidak double)**
   - **Query**

```
--soal 2
SELECT COUNT(DISTINCT id_film) AS jumlah_film_diputar
FROM JADWAL;
```

   - **Output**

	JUMLAH_FILM_DIPUTAR
1	4

3. **Member nonton > 2,3,4,5 kali**
   - **Query**

```
--soal 3
SELECT id_member, COUNT(*) AS jumlah_nonton
FROM TRANSAKSI
GROUP BY id_member;
```

   - **Output**

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.0

	ID_MEMBER	JUMLAH_NONTON
1	MM0111	3
2	MM0112	1
3	MM0113	1
4	MM0114	1
5	MM0115	1
6	MM0116	1
7	MM0117	1
8	MM0118	1
9	MM0119	1
10	MM0120	1

4. **Member termuda + jumlah tontonan**
   - **Query**

```
--soal 4
SELECT m.id_member, m.nama, COUNT(t.id_jadwal) AS jumlah_nonton
FROM MEMBER m
JOIN TRANSAKSI t ON m.id_member = t.id_member
WHERE m.tgl_lahir = (
    SELECT MAX(tgl_lahir) FROM MEMBER
)
GROUP BY m.id_member, m.nama;
```

   - **Output**

Script Output | Query Result

SQL | All Rows Fetched: 1 in 0.238 seconds

	ID_MEMBER	NAMA	JUMLAH_NONTON
1	MM0119	Dono	1

5. **Jumlah member lahir tiap bulan**
   - **Query**

```
--soal 5
SELECT TO_CHAR(tgl_lahir, 'MONTH') AS bulan,
       COUNT(*) AS jumlah_member
FROM MEMBER
GROUP BY TO_CHAR(tgl_lahir, 'MONTH')
ORDER BY bulan;
```

   - **Output**

Script Output | Query Result

SQL | All Rows Fetched: 7 in 0.01

	BULAN	JUMLAH_MEMBER
1	APRIL	1
2	FEBRUARY	1
3	JANUARY	1
4	JULY	4
5	JUNE	1
6	NOVEMBER	1
7	SEPTEMBER	1

6. **Jumlah jadwal tayang per film (periode aktif)**
   - **Query**

```
--soal 6
SELECT f.id_film, f.judul, COUNT(*) AS jumlah_jadwal
FROM JADWAL j
JOIN FILM f ON j.id_film = f.id_film
WHERE DATE '2019-06-15' BETWEEN j.periode_start AND j.periode_end
GROUP BY f.id_film, f.judul
ORDER BY f.id_film;
```

   - **Output**

Script Output | Query Result

SQL | All Rows Fetched: 4 in 0.025 seconds

	ID_FILM	JUDUL	JUMLAH_JADWAL
1	F0101	Keluarga Cemara	4
2	F0102	Habibie Ainun 3	4
3	F0103	Taufiq	1
4	F0104	Buya Hamka	1

Buatlah table perbedaan antara Agregate Function dan Scalar Function. Berikan penjelasan dan contoh nya.

Aspek	Aggregate Function	Scalar Function
Penggunaan	Mengolah banyak data (1 kolom)	Mengolah 1 data per baris
Hasil	1 nilai dari banyak data	1 nilai dari 1 data
Biasanya dipakai dengan	GROUP BY	SELECT biasa
Contoh hasil	Total, rata-rata	Manipulasi teks/angka

Contoh fungsi	COUNT, SUM, AVG, MAX, MIN	UPPER, LOWER, LENGTH, ROUND

Daftar Fungsi

Aggregate Function

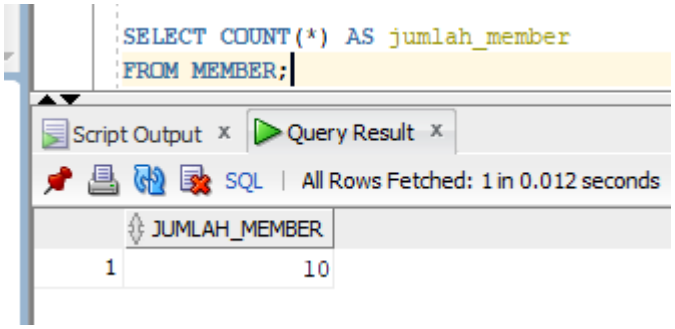
- COUNT() → menghitung jumlah data
- SUM() → menjumlahkan
- AVG() → rata-rata
- MAX() → nilai terbesar

Scalar Function

- UPPER() → huruf besar
- LOWER() → huruf kecil
- LENGTH() → panjang karakter
- ROUND() → pembulatan angka
- SUBSTR() → mengambil sebagian teks

Contoh Query

- Contoh Aggregate Function



- Contoh Aggregate + GROUP BY

```
SELECT id_member, COUNT(*) AS jumlah_nonton
FROM TRANSAKSI
GROUP BY id_member;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.004 seconds

	ID_MEMBER	JUMLAH_NONTON
1	MM0111	3
2	MM0112	1
3	MM0113	1
4	MM0114	1
5	MM0115	1
6	MM0116	1
7	MM0117	1
8	MM0118	1
9	MM0119	1
10	MM0120	1

- Contoh Scalar Function

```
SELECT UPPER(nama) AS nama_besar
FROM MEMBER;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.004 secor

	NAMA_BESAR
1	ANTO
2	BUDI
3	ARI
4	RAHMI
5	FAHMI
6	RUSLI
7	DONI
8	TATI
9	DONO
10	JOKO